Summary in Graph

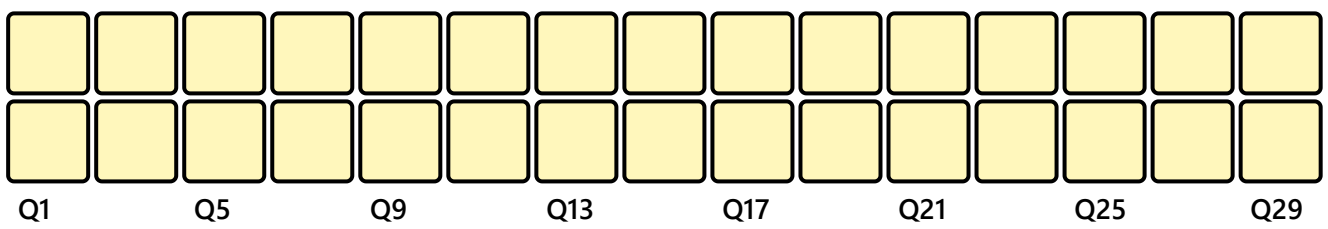
Exam Summary (GO Classes CS Test Series 2026 | Compiler Design | Subject Wise Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (15 + 15)
Total Marks:	45 (15 + 30)
Exam Duration:	75 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (34) out of 342 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Inherited attributes are attributes at a node whose dependency is

A. Restricted to siblings

B. restricted to parent

C. both A & B

D. neither A nor B

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Compiler Design

Consider the following intermediate program segment in three address code:

$$\begin{aligned} u &= v * w \\ w &= u + w \\ x &= u / w \\ v &= u - x \end{aligned}$$

The minimum number of total variables required to convert the above code segment to static single assignment form is ?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

A compiler detects:

A. errors that may occur during program execution

B. errors in the results generated by the program

C. syntax errors in the program

D. arithmetic errors

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Consider the following statements about the grammars used for syntactic analysis:

I. The $\mathrm{LL}(1)$ class does not accept grammars with productions that have direct left recursion (e.g., $\mathrm{L} \rightarrow \mathrm{La}$), but it accepts grammars with indirect left recursion (e.g., $\mathrm{L} \rightarrow \mathrm{Ra}, \mathrm{R} \rightarrow \mathrm{Lb}$).

II. The $\mathrm{LR}(1)$ grammar class recognizes the same class of grammars as $\mathrm{LALR}(1)$.

III. The $\mathrm{SLR}(1)$ grammar class recognizes a larger class of grammars than $\mathrm{LR}(0)$.

Select the correct statement:

A. Statements I, II and III are true

B. Only statements I and III are true

C. Only statement III is true

D. Only statements II and III are true

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Given the following grammar, where ε represents the empty string and $\$$ represents an end-of-input marker:

$$\begin{aligned}
\mathrm{S} &\rightarrow \mathrm{ABCD} \\
\mathrm{A} &\rightarrow \mathrm{a} \mid \varepsilon \\
\mathrm{B} &\rightarrow \mathrm{a} \mid \varepsilon \\
\mathrm{C} &\rightarrow \mathrm{c} \mid \varepsilon \\
\mathrm{D} &\rightarrow \mathrm{S} \mid \mathrm{c} \mid \varepsilon
\end{aligned}$$

It is incorrect to state that:

A. The set $\mathrm{FIRST}(\mathrm{A}) = \{\mathrm{a}, \varepsilon\}$

B. The set $\mathrm{FIRST}(\mathrm{D})$ is equal to the set $\mathrm{FIRST}(\mathrm{S})$

C. The set $\mathrm{FOLLOW}(\mathrm{A}) = \{\mathrm{a}, \mathrm{c}, \$\}$

D. The set $\mathrm{FOLLOW}(\mathrm{B}) = \{\mathrm{c}, \$\}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Analyze the following statements about recursive descent parsers.

I. They are easy *parsers* to implement for carefully designed languages, but they usually require transformations in grammars originally presented in BNF.

II. One of the main problems with this type of *parser* is the need for backtracking in alternatives, which can be solved by using a recursive predictive *parser*.

III. To avoid the problems of the recursive descent *parser*, we can perform $\mathrm{TOP\text{-}DOWN}$ analysis using a non-recursive predictive *parser*, or tabular predictive *parser*. The tabular predictive *parser* uses a table based on the FIRST and FOLLOW sets to decide which production to apply the input.

The analysis allows us to conclude that

A. only statement II is correct.

B. only statement III is correct.

C. only statements I and II are correct.

D. All three statements are correct.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type Award: 1 Penalty: 0.33 Compiler Design

Lexical analysis is usually implemented based on:

A. Regular grammar

B. Context-free grammar

C. Context-sensitive grammar

D. Unrestricted grammar

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #8 Multiple Choice Type Award: 1 Penalty: 0.33 Compiler Design

In a programming language source text, the compiler identifies the grammatical function of words and checks the grammatical structure of commands and their meanings. The architectural components of a compiler that perform these activities are, respectively,

A. lexical analyzer, semantic analyzer, intermediate code optimizer.

B. lexical analyzer, syntactic analyzer, semantic analyzer.

C. syntactic analyzer, code generator, semantic analyzer.

D. semantic analyzer, intermediate code generator, intermediate code optimizer.

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #9 Multiple Choice Type Award: 1 Penalty: 0.33 Compiler Design

What is the class of deterministic, bottom-up syntactic parsing method that processes the sequence of symbols from left to right?

A. LL

B. LR

C. Recursive Descent Parser

D. None

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #10 Multiple Choice Type Award: 1 Penalty: 0.33 Compiler Design

Analyse the following statements about the analysis phase (Front-End) of a compiler.

I. The use of a floating-point variable to index an array causes an error that is generally detected during semantic analysis.

II. Unbalanced parentheses are an error that is generally detected by lexical analysis, since this phase reads the source file and translates it into a sequence of lexical symbols, or tokens.

III. For TOP-DOWN syntactic analysis using the shift-and-reduce method, it is necessary to rewrite the grammar by eliminating all left recursion.

The analysis allows us to conclude that:

A. all statements are incorrect.

B. only statement II is incorrect.

C. only statements I and II are incorrect.

D. only statements II and III are incorrect.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Regarding basic blocks, consider the following statements.

I. The first statement can be the target of a conditional branch statement.

II. The execution flow can start between two statements in a block.

III. The execution flow can be interrupted in the middle of the block.

IV. They are used in the construction of the control flow graph.

Select the correct alternative.

A. Only statements I and II are correct.

B. Only statements I and IV are correct.

C. Only statements III and IV are correct.

D. Only statements I, II, and III are correct.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Consider the following code snippet.

$a := 0$

$b := a + 1$

$c := c + b$

$a := b * 2$

Select the alternative that correctly shows the variables that will be alive at the beginning of this code snippet.

A. b

B. b, c

C. c

D. a, b, c

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 1 | Penalty: 0 | Compiler Design

Consider the following basic block :

$$
\begin{aligned}
&// \text{ live } \{?\} \\
&a = b + 1 \\
&c = c - a \\
&b = d + b \\
&e = e - c \\
&f = f + e \\
\\
&// \text{ live } \{e\}
\end{aligned}
$$

At the bottom of the basic block, variable e is live. What will be the variables that will be live at the beginning of the basic block i.e. what belongs to $\text{live } \{?\}$?

A. a

B. b

C. c

D. d

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Let there be two grammars $G_1 \& G_2$. G_1 is left factored and free from left recursion, G_2 is not left factored and also not free from left recursion. Choose the correct option about the two grammars.

A. G_1 is an ambiguous grammar but G_2 is unambiguous.

B. G_1 is an unambiguous grammar but G_2 is ambiguous.

C. G_1 is an ambiguous grammar but G_2 may or may not be ambiguous.

D. G_1 and G_2 may or may not be ambiguous grammars.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Match the elements of list-II with the appropriate category of error given in list- I.

	LIST-I		LIST-II
(P)	Lexical	(1)	An arithmetic expression with unbalanced parenthesis
(Q)	Syntactic	(2)	An infinitely recursive call
(R)	Semantic	(3)	An operator applied to an incompatible operand
(S)	Logical	(4)	Input is not matching with any of the patterns defined

A. $\mathrm{p}-4, \mathrm{q}-3, \mathrm{R}-1,\ \mathrm{s}-2$

B. $\mathrm{p}-2, \mathrm{q}-1, \mathrm{R}-3,\ \mathrm{s}-4$

C. $\mathrm{p}-4, \mathrm{q}-1, \mathrm{R}-3,\ \mathrm{s}-2$

D. $\mathrm{p}-2, \mathrm{q}-3, \mathrm{R}-1,\ \mathrm{s}-4$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Consider the syntax directed translation scheme given below:

$$\begin{array}{ll} \mathrm{S} \rightarrow \mathrm{L} & \mathrm{S.val} = \mathrm{L.val} \\ \mathrm{L} \rightarrow \mathrm{L}_1\ \mathrm{B} & \mathrm{L.val} = \mathrm{L}_1.\mathrm{val} * 2 + \mathrm{B.val} \\ \mathrm{L} \rightarrow \mathrm{B} & \mathrm{L.val} = \mathrm{B.val} \\ \mathrm{B} \rightarrow 0 & \mathrm{B.val} = 0 \\ \mathrm{B} \rightarrow 1 & \mathrm{B.val} = 1 \end{array}$$

If the input string is 1101 , then the value of attribute of starting symbol is

A. 5

B. 13

C. 110

D. 20

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Which of the following statements regarding syntactic analysis is INCORRECT?

A. LL grammars can describe more languages than LR grammars.

B. Recursive-descent parsers are simpler to implement than bottom-up (reduction-based) parsers.

C. One of the differences among the various bottom-up parsing algorithms is the way the handle is identified on the stack.

D. In any right-sentential form, the leftmost substring that matches the body of some production is a handle.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

What is the value of the attribute E.val after parsing the expression " $4/2/2$ " using the following translation scheme?

$$\mathrm{E} \rightarrow \mathrm{T}/\mathrm{E}_1\{\mathrm{E.val} = \mathrm{T.val}/\mathrm{E1.val}\}$$

$$\mathrm{E} \rightarrow \mathrm{T}\{\mathrm{E.val} = \mathrm{T.val}\}$$

$$\mathrm{T} \rightarrow \mathrm{digit}\ \{\mathrm{T.val} = \mathrm{val}(\mathrm{digit})\}$$

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

With respect to the grammar G below,

$$\begin{aligned}
&\mathrm{S} \rightarrow \mathrm{a\ A\ a} \\
&\mathrm{S} \rightarrow \mathrm{b\ A\ b} \\
&\mathrm{A} \rightarrow \mathrm{b} \\
&\mathrm{A} \rightarrow \mathrm{epsilon}
\end{aligned}$$

Consider the following statements :

I. G is $\mathrm{SLR}(1)$.

II. G is $\mathrm{LL}(1)$.

III. It is sensitive to context.

It is correct to say that :

A. Only I is true.

B. Only II is true.

C. Only III is true.

D. Only I and III are true.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Consider a production belonging to a grammar G given by :

$$\mathrm{L} \rightarrow \mathrm{LaS} \mid \mathrm{S}$$

Select the option below that, by replacing this production, eliminates left recursion, creating an equivalent grammar :

A. $\mathrm{L} \rightarrow \mathrm{RS}$
$\mathrm{R} \rightarrow \mathrm{aSR} \mid \varepsilon$

B. $\mathrm{L} \rightarrow \mathrm{SR}$
$\mathrm{R} \rightarrow \mathrm{aSR} \mid \varepsilon$

C. $\mathrm{L} \rightarrow \mathrm{SR}$
$\mathrm{R} \rightarrow \mathrm{SaR} \mid \varepsilon$

D. $\mathrm{L} \rightarrow \mathrm{SaR}$
$\mathrm{R} \rightarrow \mathrm{SaR} \mid \varepsilon$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Consider the following grammar for expressions.

$$\begin{aligned} S &\rightarrow E\$ \\ E &\rightarrow E + T \\ E &\rightarrow T \\ T &\rightarrow T * F \\ T &\rightarrow F \\ F &\rightarrow id \\ F &\rightarrow (E) \end{aligned}$$

With respect to this grammar, consider the statements below.

I. The grammar is $\mathrm{LL}(1)$.

II. The operator $+$ has higher precedence than the operator $*$.

III. It is not possible to construct a recursive-descent parser for the grammar.

IV. The terminals $+,\ *,\ ),$ and $\$$ belong to the FOLLOW set of F.

Select the correct option.

A. Only statements I and II are correct.

B. Only statements I and IV are correct.

C. Only statements III and IV are correct.

D. Only statements I, II, and III are correct.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 06sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Consider the following grammar.

$$\begin{aligned} \mathrm{E} &\rightarrow \mathrm{num} \\ \mathrm{E} &\rightarrow \mathrm{E} + \mathrm{E} \\ \mathrm{E} &\rightarrow \mathrm{E} - \mathrm{E} \end{aligned}$$

Based on this grammar, assign T (true) or F (false) to the following statements.

I. It is ambiguous.

II. It is $\mathrm{LL}(1)$.

III. It is $\mathrm{LR}(1)$.

IV. It is SLR.

V. It has left recursion.

Select the option that contains, from top to bottom, the correct sequence.

A. T, T, F, F, T

B. T, F, T, T, F

C. T, F, F, F, T

D. F, T, T, F, F

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

An intermediate representation of the source program can be generated by transforming the derivation tree into a segment of code. With respect to the intermediate code generation phase of a compiler, which of the following alternatives is INCORRECT?

A. With a properly defined intermediate representation, a compiler for any language i and any machine j can then be built by combining the front end for language i with the back end for machine j.
B. Syntax trees and three-address code are some of the possible forms of intermediate representation.
C. High level languages, such as C , can be used as a form of intermediate representation.
D. In intermediate code generation, tasks such as instruction selection, register allocation and assignment, and instruction scheduling are performed, which depend on knowledge of the target machine for which the object code will be generated.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

The main task of a lexical analyzer is to read the characters from the source program input, group them into lexemes, and generate a sequence of tokens that will be sent to the syntactic analyzer. Regarding the lexical analyzer, analyze the statements below:

I. In addition to identifying lexemes, other tasks can be performed by this analyzer, such as removing comments and whitespaces and associating error messages with lines in the source program.

II. A token is a pair consisting of a token name (or token type), which identifies the category of the lexeme (such as identifier, keyword, or number), and an attribute value, which provides additional information about the lexeme (for example, a pointer to the symbol table entry).

III. Regular expressions are notations used to specify lexeme patterns.

IV. In lexical analysis, an intermediate tree-like representation is created. This represents the grammatical structure of the token sequence.

Which ones are correct?

A. Only III.

B. Only I, III.

C. Only IV.

D. Only I, II, III.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

The following syntax-directed translation scheme translates a language with terminals $\mathrm{a}, \mathrm{b}, \mathrm{c}$, and d into a language whose terminals are $1, 2, 3, 4, 5, 6$. Using a bottom-up parser that executes the actions in parentheses immediately after reducing the corresponding rule, what is the result of translating "aaadbc"?

$$\begin{array}{ll} \mathrm{S} \rightarrow \mathrm{AS} & \{\text{print "1"}\} \\ \mathrm{S} \rightarrow \mathrm{B} & \{\text{print "2"}\} \\ \mathrm{A} \rightarrow \mathrm{a} & \{\text{print "3"}\} \\ \mathrm{B} \rightarrow \mathrm{bC} & \{\text{print "4"}\} \\ \mathrm{B} \rightarrow \mathrm{dB} & \{\text{print "5"}\} \\ \mathrm{C} \rightarrow \mathrm{c} & \{\text{print "6"}\} \end{array}$$

A. 1313132546

B. 1113332546

C. 3336452111

D. 6452111333

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

In the process of generating executable code (in machine language) from a source program written in a high-level language (for example, C), the original program undergoes transformations and analyses that are carried out in several phases. In a simplified way, these can be divided into the following eight (8) phases, listed below in alphabetical order:

A. Register Allocation

B. Lexical Analysis

C. Syntactic Analysis

D. Assembly Code Generation

E. Link Editing (Linking)

F. Assembly

G. Machine Independent Code Optimization

H. Type and Symbol Checking

During the process of generating executable code from the source code, in what order can these phases be executed?

A. B C H G A D F E

B. C B H G A D F E

C. B C H G A D E F

D. B H C G A D F E

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Observe the following grammar:

$$\begin{aligned}
\text{DECL} &\rightarrow \text{IF} \mid \text{a} \\
\text{IF} &\rightarrow \text{if}\,(\,\text{EXP}\,)\ \text{DECL}\ \text{ELSE} \\
\text{ELSE} &\rightarrow \text{else DECL} \mid \varepsilon \\
\text{EXP} &\rightarrow 1 \mid 0
\end{aligned}$$

Where $ is the symbol representing the end of the file, it is ***CORRECT*** to state that

A. $\text{FIRST(DECL)} = \{\text{if}, \text{a}\}$ and $\text{FOLLOW(DECL} = \{\$\}$

B. $\text{FOLLOW(DECL)} = \text{FOLLOW(IF)} = \text{FOLLOW(ELSE)} = \{\$, \text{else}\}$

C. $\text{FIRST(DECL)} = \text{FIRST(IF)} = \{\text{if}\}$

D. $\text{FIRST(ELSE)} = \{\text{else}\}$ and $\text{FOLLOW(ELSE)} = \backslash\{\$\backslash\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Consider the following grammar. (Assume $\$$ is the end marker.)

$$A \rightarrow a$$
$$A \rightarrow C\,B\,A$$
$$B \rightarrow \varepsilon \mid b$$
$$C \rightarrow B \mid c$$

Regarding this grammar, select the correct alternative.

A. The FIRST set of B is $\{b, c\}$.
B. The FIRST set of C is $\{a, b\}$.
C. The FOLLOW set of A is $\{\$\}$.
D. The FOLLOW set of C is empty.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

For the grammar below, what is the set of terminals that can appear as the first terminal after the nonterminal A, in some sentential form generated by the grammar below, where ε represents the empty string?

$\text{S} \rightarrow \text{ABCDd}$

$\text{A} \rightarrow \text{aA} \mid \varepsilon$

$\text{B} \rightarrow \text{bC} \mid \varepsilon$

$\text{C} \rightarrow \text{cD} \mid \varepsilon$

$\text{D} \rightarrow \text{e}$

A. $\{\text{d}\}$

B. $\{\text{b}\}$

C. $\{\text{b}, \text{c}, \text{e}\}$

D. $\{\text{b}, \text{c}, \text{d}, \text{e}\}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following 3 Address Intermediate Code.

1. $P := 0$

2. $I := 1$

3. $P := P + I$

4. $IF\ P <= 60\ \text{GOTO}\ (7)$

5. $P := 0$

6. $I := 5$

7. $T1 := I * 2$

8. $I := T1 + 1$

9. $IF\ I <= 20\ \text{GOTO}\ (3)$

10. $K := P * 3$

Let x be the number of Nodes in the Control Flow Graph (Excluding the two special nodes ENTER and EXIT) for the above IR and let y be the number of edges in the CFG then $x + y$ is _____ ?

Your Answer: | **Correct Answer: 11** | Not Attempted | Time taken: 00min 00sec | Discuss

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